

SecureIN Session Plan

Day	Hr	Topic	Sub-Topic	Hands-On
Day 1	1	TCP/IP Deep Dive	TCP vs UDP, 3-way handshake, ports & sockets, OSI vs TCP/IP mapping	Use netstat, ss to identify active connections and ports
Day 1	2	DNS + HTTP/HTTPS	DNS resolution flow, HTTP lifecycle, HTTPS (TLS basics), headers	Use nslookup, dig, inspect request/response in browser DevTools
Day 1	3	Routing & Switching	ARP, MAC table, L2 vs L3, subnetting tricks, NAT	Solve subnetting problems, identify IP classes
Day 1	4	Network Security Basics	Firewall, IDS/IPS, VPN, MITM, DDoS, DNS spoofing	Wireshark demo: capture packets & analyze HTTP vs HTTPS
Day 1	5	Interview + Scenario Practice	End-to-end request flow, debugging network issues, output-based question	Analyze “what happens when you open a website” + MCQs
Day 2	1	DevOps Fundamentals	DevOps lifecycle, CI/CD overview, SDLC vs DevOps	Map lifecycle of a real project (discussion-based activity)
Day 2	2	Git + CI/CD	Git workflow (clone, commit, push), branching, CI/CD basics	Push code to GitHub, simulate simple pipeline
Day 2	3	Cloud Basics	IaaS, PaaS, SaaS, deployment flow, load balancing	Deploy a simple app using Render/Vercel
Day 2	4	DevSecOps	Secure coding basics, OWASP Top 10, vulnerabilities overview	Identify issues in sample vulnerable code
Day 2	5	Mini Project + Interview Prep	Deploy + secure application, common interview questions	End-to-end: push → deploy → find security gaps